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# Laser strengthening of CP65+5%MΓ abrasive material for gears of oil pumps

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**Abstract.** The article is devoted to the study of the structure and properties of the surface layer of СП65+5%МГ abrasive steel subjected to laser processing and alloying. Here, the effects of laser processing and alloying on surface layer hardness, corrosion resistance and porosity were evaluated. It has been found that, compared to baked nitrocemented and unfused laser lazing, alloying with chromium, boron carbide composition, during continuous laser processing, the low level of radiation does not strongly affect the structure of the material and cracks do not occur. The highest microhardness was achieved in the lower values of the surface layer.

**Key words:** Oil pump, gear wheel, abrasive steel, laser processing, strengthening, tempering, melting, alloying, structure, microhardness.

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## 1. Introduction

Mechanical and operational properties of steel and alloys in machine building and other fields quite different methods are used to raise it. Addition of various elements, plastic deformation, thermal, chemical-thermal processing, etc. are among these methods [1]. In order to improve the more efficient use of low-alloyed scrap steels, determining the regularity and relationship between their internal structure and mechanical properties is one of the important issues of materials science. The main condition is the transformation of the structure of the steel and its transformation into powder to change its various properties [2]. It has been found that it is recommended to make a relatively simple symmetrical shape, small mass and size product from rubbing materials. The amount of mechanical processing used in the manufacture of gears is reduced by means of scrap metallurgy. It also reduces the labor capacity and the price of the product [3]. One of the new research directions is to apply laser tabulation and alloying to oil pump gears by reducing the number and amount of alloying elements, taking into account the working environment. In short, Cu, Ni, Cr, etc. instead, surface strengthening by laser processing by incorporating copper-plated graphite in carbon steel can be considered one of the most pressing issues.

## 2. Experimental detail

СП65+5%МГ grinding, chromium grinding of structural steel and boron The structure and phase composition of the surface layer after laser alloying with the help of carbide composition was studied. The gear wheel detail made of the investigated SP65+5%MΓ abrasive material is used in oil pumps and was accepted as a research object. The working drawing of gear pump and gear wheel is shown in figure 1 and 2.

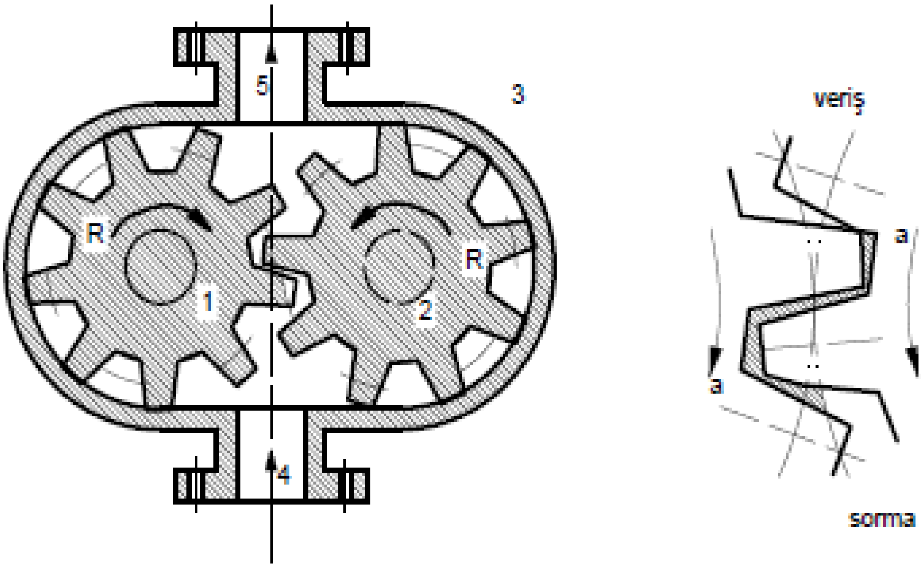



Figure 1. Working diagram of a gear pump

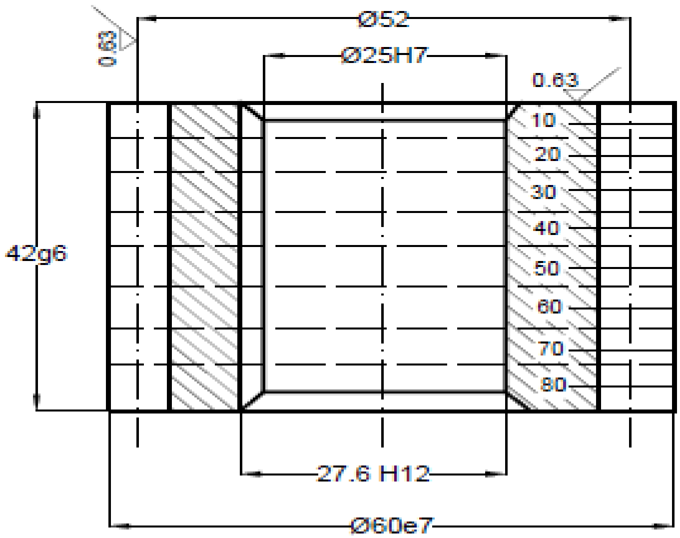
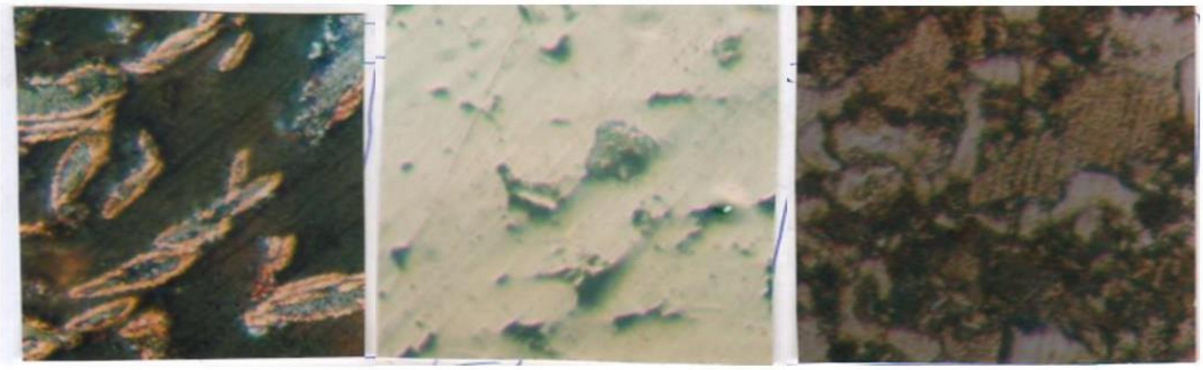



Figure 2. Working drawing of a gear wheel

Laser processing on the IXUN Lasertechnik Gmbh device (energy pulse 5-9 C, beam beam diameter 1.0-1.2 mm, surface spot coverage 50-70%) and ЛТН (irradiation power 90-132 W, beam beam diameter 1.0÷1.2, displacement speed 180-240 mm/min), structural examination in PMI OLMPUC brand microscope, phase composition Cobalt monochromatization is determined by Kα radiation in a DPOH-3 X-ray diffractometer. The abrasive construction material subjected to laser treatment [4] was reinforced with SP65 slag content and SP65+5%MG with copper-plated graphite addition [5]. Figure 3 shows the copper-coated graphite particle shape and the microstructure of the sintered SP65 steel.



a b c

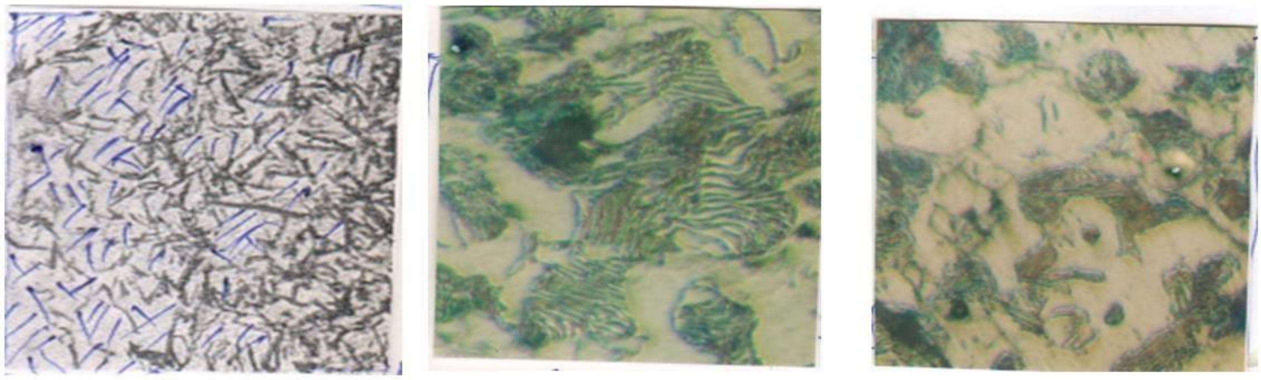
Şəkil 3. Mislənmiş qrafitin və СП65 poladın mikrostrukturu

a-mislənmiş qrafit; b,c-bişirilmiş СП65, aşındırılmamış; a-aşındırılmış (C)

Tribotechnical characteristics were carried out with the help of a special design [6] on the СМЦ-2 machine at constant loading according to the 《disc-pad》 scheme, and the hardness was performed on the ПМТ-3 hardness tester. The initial state of the surface of Ovuntu structural steel is porous with a hardness of 250 H. The phase composition in the initial state consists of Feα, Feγ, Fe3C, C. Conventional chemical thermal treatment cannot obtain uniform hardened layer on the surface of steel. After nitrocementing, a high-hardness layer (600-700H) around the pore, which is not located in the surface layer, but at a considerable distance from the surface. On the other hand, the hardness of the original material on the surface is manifested in many areas. This proves that carbon and nitrogen displacement during nitrocementing of abrasive construction material is pore-wise, so it goes to great depth with uneven performance.

In the study, the phase composition on the surface of SP65+5%МГ steel is Feα, Feγ, Fe3C, FeN, Fe3N, C. At this time, first of all, nitrogen penetrates the metal around the distance. During nitrocementing, there is end-to-end penetration in the cross-section, which is not very detailed, which indicates that the metal is soft. It was found that the surface layer of SP65+5%МГ structural steel consists of structural elements - ferrite, cementite and free carbon in the initial state. After nitrocementitizing, nitrogenous martensite and iron nitride are formed. In the initial state of steel, graphite is observed in the steel structure, and during laser processing, graphite is partially formed, while the rest dissolves and stabilizes austenite (picture 4). Melting the surface with a laser beam changes the phase composition of the surface. Martensite, cementite and a small amount of residual austenite are formed in the structure.

The microstructure of the surface layer of SP65+5%МГ steel after laser melting and alloying consists of two zones: the melting zone and the associated thermally affected zone (Fig. 5). The melting zone behaves as a poorly eroded layer and is formed as a result of quenching from the liquid state, consisting of fine-grained martensite with a microhardness of 750H. In this regard, the porosity of the abrasive steel is not involved. In a very limited time, the material is liquid in the surface layer and strongly convective processes do not have time to release the gas in the primary material in the melting bath, and it accumulates in various sizes as a spherical pore near the bottom of the melting zone. The phase composition is Feα, Feγ, Fe3C (Fig. 5,a).



a b c

Şəkil 4. СП65+5%МГ poladın səthinin əriməmiş emalı (lazerlə tablandırma)

a-xırda iynəvari martensit (750H); b-perlit (500H); c-ferrit+perlit (300H)

During tabulation, a thermally affected zone is formed from the austenite area. In such a case, the microhardness gradually decreases with the increase of ferrite from 750H. The material heats up quickly and cools down quickly, so it does not have time to homogenize. As a result of the thermal effect, traces of pearlite appear and the formation of fine-grained martensite begins in that zone. Residual austenite formation occurs during this formation. The phase composition Feα, Feγ, Fe3C, C (traces) is observed (Fig. 4, b).

During laser alloying of steel with chromium and boron carbide, reinforcing FeB, Fe2B and Cr7C3, Cr2C6 phases and B4C lubricating insoluble particles are formed in the structure of the laser processing layer in martensite and cementite (phase composition in SP65+5% MG steel in laser alloying is Feα, Feγ, Fe3C, FeB, Fe2B, Cr7C3, Cr2C6), the strengthening phase in the structure as a result of its increase, an increase in hardness of 1400-1500H is observed (table 1). As a result, the saturation of solid solutions formed by the alloying elements and the large temperature difference in the alloying zone will lead to a considerable surface hardness. It should also be noted that during processing, the stresses on the surface of the pulsed laser radiation form a crack and continue in the direction of heat transfer. During continuous laser processing, a weak level of radiation does not strongly affect the structure of the material, and cracks do not appear (Fig. 5,c).

Tribotechnical properties were studied in SP65+5%МГ steel in the initial composition and after nitrocementitizing, laser tempering and laser alloying (table 2).

**Table 1. Variation of microhardness across the cross-section of the surface layer of SP65+5%МГ steel (Thickness of the layer from the surface-h)**

| Processing | Microhardness, H | | | | |
|---|---|---|---|---|---|
| | The thickness of the layer from the surface-h, μm | | | | |
| | 100 | 200 | 300 | 400 | 500 |
| Primary material | | | | 300 | - |
| After microsemenetitization | 750 | 620 | 360 | 300 | - |
| Laser melting | 750 | 530 | 350 | 300 | - |
| Laser alloying (with Cr+B4C comp.) | 1350 | 1000 | 625 | 300 | 310 |

a b c

Şəkil 5. СП65+5%МГ poladın lazerlə emaldan sonra səth qatının mikrosrukturu

a-nitrosementitləmə; b-lazerlə əritmə; c-Cr+$B_4C$ kompozisiya ilə legirləmə

**Table 2. Dependence of tribotechnical properties in processing methods**

| Processing | f | İ , mm/km | $t_{iş}$ , dəq |
|---|---|---|---|
| Initial composition, cooked, СП65+3%МГ | 0,144 | 1,429 | 4,9 |
| Non-melting laser lamination of the surface (《IXUN Lasertechnik Gmbh》) | 0,140 | 0,403 | 3,9 |
| Alloying with Cr+B4C composition (《IXUN Lasertechnik Gmbh》) | 0,140 | 0,016 | 2,1 |
| Nirtosementitization | 0,135 | 0,039 | 3,4 |

Marking: f - coefficient of friction, I - wear intensity, tooth working time

**3. Conclusion**

It has been shown that conventional chemical heat treatment produces a high hardness layer on the surface of the steel, but not located in the surface layer but far from the surface. It was found that after nitrocementitizing, martensite and iron nitride are formed with nitrogen. Laser melting of the surface changes the spatial composition of the surface and martensite, cementite and a small amount of residual austenite are formed in the structure. It is clear from the study that during laser alloying of steel with chromium and boron carbide, martensite and cementite strengthening phases and B4C lubricant insoluble particles are formed in the laser treatment layer in the structure. It was found that during continuous laser processing, a weak level of radiation does not strongly affect the structure of the material and prevents the formation of cracks. Research results show that the best tribotechnical properties in СП65+5%МГ steel subjected to various treatments were obtained in laser alloyed treatment with B4C+Cr composition.

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